INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

TARGET SYSTEM, EXTERNAL TOW, GUNNERY, A/A3-7-U-15

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an externally mounted gunnery tow target system, designated A/A37U-15.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 or 4 of this specification. This section does not include documents specified in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-8591 - Airborne Stores, Suspension Equipment, and Aircraft - Store Interface (Carriage Phase); General Design Criteria For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: Ogden ALC/LIRF, 6041 Gum Lane, Hill Air Force Base, Utah 84056-5826, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6920

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MIL-T-9918 - Tow Target, Aerial Dart Gunnery Type TDU-10B

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Test Method Standard for Environmental Engineering Considerations and Laboratory Tests

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, Pennsylvania 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues are those cited in the solicitation.

DRAWINGS

AIR FORCE

61J10536	Target System, External Tow, Gunnery A/A37U-15
61J10905	Rewind Stand, THU-4/E

(Copies of drawings are available from the Ogden Air Logistics Center, OO-ALC\TIEDD, 6032 Fir Avenue, Hill AFB, Utah 84056-5609.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of the documents which have been adopted by the DoD are those contained in the specific issue of the DoDISS cited in the solicitation. Unless otherwise specified, the documents not contained in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Anderson, Greenwood & Company

29-0402	Tow Reel, Model 29C2
29B0495	Adapter Kit, F-4 Aircraft
29D0768	Installation Kit, F-5 Aircraft

(Application for copies of non-government drawings should be addressed to the procuring activity or as directed by the contracting officer.)

2.4 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. A conflict between this document and its related associated specifications or specification sheets should be brought to the attention of the issuer of the solicitation.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.2.

3.2 Design. The target system shall conform to Drawing 61J10536.

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3.2.1 Components. The target system shall consist of the following major components:

ITEM NO	DESCRIPTION	QUANTITY
1	Pod, Tow Reel RMU-10/A	1
2	Boom-Target Adapter	1
3	Launcher-Target Boom	1
4	Anderson Greenwood Tow Reel, Model 29C2	1

3.2.1.1 Tow Reel Pod. The tow reel pod shall be designed to minimize aerodynamic drag in the operating environment. The design and location of the tow reel brake drum shall be such as to prevent overheating of the brake assembly.

3.2.1.1.1 Cable capacity. The capacity of the tow reel spool shall be a minimum of 2200 feet of 11/64-inch armored tow cable.

3.2.1.1.2 Servicing provisions. Capability shall be provided for removal of the empty spool and replacement with a prewound cable spool without removing the reel and reel housing from the aircraft.

3.2.1.2 Boom-Target Adapter. The boom-target adapter shall provide the mechanical and electrical connections between the tow reel pod and the launcher-target boom, and shall meet the requirements of Drawing 61J10536.

3.2.1.3 Launcher-Target Boom. The launcher-target boom shall provide the mechanical and electrical connections between the boom-target adapter and the target, and shall meet the requirements of Drawing 61J10536.

3.2.2 Installation kits. Installation kits for the target system shall be adaptable, without mechanical or electrical modification, to the following aircraft:

AIRCRAFT	DRAWING NO.
F-4	29B0495
F-5	29D0768

3.3 Performance characteristics. The target system (without the tow target attached) shall conform to a 1000-pound-class store in flight as specified in MIL-A-8591. This system shall be designed to perform under a maximum aerodynamic load of 4g, at Mach 1.3, 30,000 feet mean sea level (msl).

3.3.1 Operation. The target system shall comply with the following functional requirements:

a. Provide mounting and firm support of the TDU-10/B dart tow target during taxi, take-off, and flight to 350 knots indicated air speed (KIAS) at a load of 1.5g.

b. Provide smooth release and deployment of the target to the full length of the tow cable. Release and deployment shall be accomplished at a minimum of 200 KIAS at altitudes from 1,000 to 20,000 feet msl.

c. Provide secure attachment of the TDU-10/B dart tow target during aircraft towing to a maximum airspeed of 450 KIAS at 0.9 Mach with 2.5g loading.

d. Allow initiation of target recovery by severing the tow cable at airspeeds above 200 KIAS and altitudes between 1000 and 20,000 feet msl. The mechanism used to sever the cable shall include primary and back-up cable cutters.

e. Provide secure attachment of the target system without the target and cable attached during aircraft flight at speeds to Mach 1.3, at loads of 4g, and at altitudes to 30,000 feet msl.

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3.3.2 Environmental conditions. The target system shall meet the requirements of this specification under the following conditions:

a. Any altitude between zero and 30,000 feet msl.

b. Vibration incident to the target system installed on the aircraft listed in 3.2.2.

3.3.3 Acceleration loads. The target system shall withstand acceleration loads typical of normal aircraft installations for a 1000-pound class store in flight, as specified in MIL-A-8591.

3.4 Supportability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.4.1 Accessibility. The tow target system shall afford accessibility to installed components to facilitate ease of maintenance.

3.4.2 Identification. Equipment, assemblies, containers, and parts shall be marked for identification. The nomenclature and stock number to be included on the equipment nameplate shall be as follows:

Target System, External Tow, Gunnery A/A37U-15
Stock Number 6920-00-056-6036

3.5 Color and finish. Color and finish shall be in accordance with applicable drawings.

3.6 Safety. The tow target system shall be functional without sharp edges, burrs, or protrusions that might injure personnel.

3.7 Workmanship. The target system, including all parts and equipment, shall be constructed in accordance with commonly accepted industrial workmanship standards.

3.8 Government-furnished property. When specified in the contract or purchase order (see 6.3), the Government will furnish the following to the contractor upon request, for use in fabrication of the tow target system:

STOCK NO.	NOMENCLATURE
1095-00-141-9328	Rack Assembly, Bomb, Type MA-4A

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

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4.2 First article inspection.

4.2.1 First article test samples. The first article test samples shall consist of the first five target systems representative of the production equipment. Three target systems shall be used for environmental and static load tests by the contractor, and two target systems shall be delivered for inspection and flight testing by the Government, as directed by the procuring agency.

4.2.2 First article inspection requirements. First article inspection shall consist of all examinations and tests described in 4.4.

4.3 Conformance inspection. Each target system shall be subjected to the individual examinations and tests as described in 4.4.1 and 4.4.2.

4.4 Methods of inspection.

4.4.1 Examination. Each target system shall be examined for compliance with the requirements specified in 3.2, 3.3, 3.4.2, 3.5, and 3.6. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement, or presence of one or more defects preventing or lessening maximum efficiency, shall constitute cause for rejection.

4.4.2 Ground operation. Power shall be applied to the target system to determine compliance with Drawing 61J10536 bomb rack and cable cutter circuitry.

4.4.3 Environmental tests. All environmental tests will be accomplished with an empty tow reel spool. Failure of any of these tests will be cause for rejection.

4.4.3.1 Vibration tests. The target system shall be mounted on the vibration table in the same manner as mounted on the applicable aircraft. The vibration procedure shall be as follows:

a. Vibrate the system in a horizontal direction with the frequency varying between 10 and 55 cycles per second (cps) at a single amplitude of 0.03 inch (0.06 inch total excursion). The frequency shall be changed uniformly from 10 to 55 cycles and returned to 10 cycles in 1 to 3 minutes. This test shall continue for 90 minutes. During this time the frequency of any and all resonant points (natural periods) shall be noted. If more than three resonant points are encountered with vibration applied along one axis, the three most severe shall be chosen for the test.

b. Vibrate the equipment for 15 minutes at each of the resonant frequencies noted in step (a) at 0.06 inch total excursion.

c. Repeat steps (a) and (b) changing the direction of vibration 90° horizontally.

d. Repeat steps (a) and (b) changing the direction of vibration to vertical.

e. Remove the equipment from the table and visually inspect for any mechanical failures. There shall be no visible sign of cracking, separation, or other physical failures.

4.4.4 Static load tests. The target system shall be mounted to a structural frame in the same manner as mounted on the applicable aircraft. Critical loading conditions as specified in MIL-A-8591 will be applied and deflection will be recorded. There shall be no permanent set, deformation, or structural damage.

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4.4.5 Flight tests.

4.4.5.1 Without target test. The target system without the target will be flown on the applicable aircraft at speeds to Mach 1.3, at loads to 4g, and at altitudes to 30,000 feet msl. The tow reel will have a full load of cable.

4.4.5.2 Target support test. A TDU-10/B dart tow target will be attached to the A/A37U-15 target tow system and the applicable aircraft will taxi, take-off, and fly to 350 KIAS at loads to 1.5g.

4.4.5.3 Target launch test. The TDU-10/B dart tow target conforming to MIL-T-9918 will be launched from the target system at a minimum of 200 KIAS at an altitude between 1,000 and 20,000 feet msl.

4.4.5.4 Tow target test. The target system will be flown with the target in the tow position to speeds of 450 KIAS and 0.9 Mach at loads of 2.5g.

4.4.5.5 Target recovery test. The target system will be used to initiate target recovery at a minimum of 200 KIAS and at an altitude between 1,000 and 20,000 feet msl.

4.4.5.6 Post-flight inspection. After completion of flight testing, the target system shall be examined and tested in accordance with 4.4.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The target tow system covered by this specification is intended to provide a means to tow, with F-4 or F-5 aircraft, a TDU-10/B Dart Gunnery Tow Target for use in gunnery training of combat crew members.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. When first article inspection is required (see 3.1).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.7.

6.4 Rights in data. The Government has rights of reproduction and disclosure in connection with the Anderson Greenwood Drawing 29-0402 covering the Model 29C2 Tow Reel, Drawing 29B0495 covering the adapter kit for the F-4 aircraft, and Drawing 29D0768 covering the installation kit for the F-5 aircraft.

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6.5 Subject term (key word) listing.

Gunnery
Tow Target
Aircraft

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.7 Ground Support Equipment. A THU-4/E Rewind Stand, Air Force Drawing 61J10905, is used in conjunction with this system as ground support equipment.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 70

Project NR 6920-0142

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-9813B	960605

3. DOCUMENT TITLE TARGET SYSTEM, EXTERNAL TOW, GUNNERY A/A3-7-U-15

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Pete Schwager, LIRBA	b. TELEPHONE *(Include Area Code)* (1) Commercial (801) 777-5603 (2) DSN 458-5603
c. ADDRESS *(Include Zip Code)* *6041 Gum Lane* *Hill Air Force Base, Utah* *84056-5826*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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